A Controlled Replication of a Self-Model Loop for Language-Model Agents: Calibration, Causal Attribution, and Closed-Loop Hysteresis

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August 30, 2026

Abstract

A recently described architecture wraps a large language model (LLM) with a self-referential inference layer: a Bayesian belief over the agent's own interaction "stance", re-injected into generation each turn, together with a hidden deliberative monologue. The originating write-ups report strong effects but do not publish the controls needed to separate the contribution of the state *content* from the presence of an instruction, an honest task baseline, or the intrinsic inertia of the estimator. We give an independent, clean-room reimplementation of the loop and subject its central claims to those controls across seven experiments. We find: (i) under a structured-placebo control, the state's *content* is causally load-bearing only when the posterior is confident, and the effect is carried most clearly by the hidden monologue; (ii) a clamped factorial confirms calibration (the stance predicts the agent's own reply behaviour) and shows the "gate" token to be a minor, non-uniform channel; (iii) on a ground-truth hidden-stance task the loop infers at 3× chance but *ties* an honest single-shot LLM baseline, so the filter buys persistence rather than accuracy— and a bare LLM is far above chance, so reporting chance as the baseline inflates the apparent gain; (iv) driving the closed loop with a ramped external push produces hysteresis (path-dependent memory) but no sharp bistable jump; a state-decoupled null run shows that roughly two thirds of the raw loop area is accumulator arithmetic, leaving a modest ($\sim$0.07 in loop area), rate-independent loop-attributable excess that grows rather than shrinks as the ramp slows; and a long-horizon basin test finds a single attractor—initial conditions remain visible for over 100 turns, but there is no bistability. Every claim tested reproduces in a "true-but-softer" form once the missing controls are added. We make no claim about consciousness or any threshold measure; the scope is strictly the loop's measurable behaviour. Code and data are released.

1 Introduction

Adding a persistent, self-referential state to an LLM agent—a belief about the agent's own stance, updated online and fed back into generation—is an appealing idea with a growing informal literature. A prominent recent instantiation (hereafter "the AC1 loop", following its author's naming) combines four ingredients: a Bayesian stance tracker; a rendered state *prefix* injected upstream of the reply; a hidden *private monologue* that deliberates before the public reply; and a second-order term modelling the interlocutor. The originating essays report large effects—for example that removing the monologue changes the output on 92–100% of measurable turns—and frame these as evidence for strong properties.

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Our interest is narrower and purely methodological. Effects of this kind can arise for reasons that have nothing to do with a self-model: any prefix that contains an imperative will change behaviour; a "baseline" set at chance will make any above-chance system look impressive; and an estimator that accumulates evidence will look path-dependent whether or not the surrounding loop is—with decay, through inertia; without decay, by construction, since an undecayed accumulator never forgets. The originating write-ups do not report the controls that separate these. This paper supplies them, on an independent reimplementation, and reports what survives.

Scope and non-claims. We study only the measurable behaviour of the loop. We make **no** claim about the Ψ-threshold, consciousness, or the wider conjecture the architecture is embedded in; those questions are neither tested nor implied here. Where a claim holds we say so; where it holds only in a softer form, or rests on an unresolved confound, we say that too.

2 The loop

The state is a Dirichlet posterior over five stance modes {exploring, evaluating, overwhelmed, asserting, disengaging}. Each turn:

1. a **sensor** (an LLM classification with confidence) reads the latest line and yields a per-mode likelihood ℓ and a scalar evidence e;
2. the tracker updates $\alpha \leftarrow \rho\,\alpha + e\,\ell$, with a mean-step cap, an α floor, and a concentration cap;
3. the posterior $p = \alpha/\kappa$ is rendered into a **prefix** (the five values, the dominant mode, a *gate* flag, and an optional per-mode *strategy* line);
4. an optional **private monologue** is generated from the prefix and prepended, unseen, to the reply's system prompt;
5. the reply is produced; the sensor reads it; the loop repeats.

The decay $\rho < 1$ gives the posterior finite memory. Our reimplementation is standard-library Python against any OpenAI-compatible endpoint; it copies no part of the original codebase and is built only from the public descriptions.

A practical note that matters for reproduction. Some reasoning models return their text in a `reasoning_content` field with an empty `content`. A sensor reading only `content` then silently returns nothing and the loop degrades *without erroring*. Our extractor falls back to `reasoning_content`; anyone reproducing this against a reasoning model must do the same or measure a broken loop.

3 Experiments

All behaviour metrics are ground-truth (word counts, question flags, matches to a known hidden label); where a blind label is needed (E1) it is produced by a separate model that never sees the prefix. Figures below are from a small number of base models and should be read as illustrative in magnitude and indicative in direction.

Clamped mode	mean words	question rate
exploring	51.0	1.00
evaluating	102.2	0.50
overwhelmed	108.0	0.83
asserting	46.0	0.50
disengaging	133.5	0.50

Table 1: E2 calibration: clamped dominant mode vs. reply behaviour (strategy on, monologue off, gate off). The mode predicts style—exploring is short and always questions; asserting is short and direct; the deliberative modes run long.

3.1 E1: structured-placebo ablation

Four arms at matched format: TRUE (production prefix), SCRAMBLED (same posterior with mode labels deranged, dominant and strategy rebuilt so the placebo is internally consistent but wrong), NUMBERS_ONLY (true posterior, strategy line stripped), NULL (no prefix). Under a *weakly* held posterior, TRUE≈SCRAMBLED on the blind-label channel and NUMBERS_ONLY≈NULL on every channel: the imperative *strategy line* carries the visible effect, not the state values. Taken alone this is deflationary—the model follows the instruction whether the state behind it is true or deranged.

3.2 E2: clamped factorial

Clamping the posterior to each mode in turn (so the design cannot collapse into a single mode) and toggling gate, monologue and strategy independently (180 cells), we find that under a *strong* clamped posterior:

- the dominant mode **predicts** reply behaviour (Table 1); calibration holds;
- NUMBERS_ONLY prefixes now produce long, unshaped replies (∼150–200 words) that differ sharply from strategy-bearing prefixes (46–133 words): the state values are *not* universally inert. This **corrects** the E1 reading—inertness was a weak-state artefact;
- the gate token moves length modestly and non-uniformly (some modes lengthen, some shorten): a real but minor secondary channel, not the driver;
- the monologue sharpens state-consistent behaviour where the state is strongest.

In a companion two-stage variant on a non-reasoning base (so the monologue is the only deliberation), TRUE/SCRAMBLED/NULL produced reply lengths of ∼131/47/79 words—opposite-signed departures from null in the state-consistent direction ($t \approx 5.2$ for TRUE vs. SCRAMBLED). Chained with the NUMBERS_ONLY≈NULL result, this attributes the causal channel to the **hidden monologue**: it is the step at which the state's content becomes behaviourally load-bearing. This reproduces the architecture's own "prefix alone is insufficient" claim, under a structured-placebo control the original does not report.

3.3 E3: hidden-stance game with an honest baseline

Each episode fixes a hidden mode; a simulator writes three short in-mode lines; three predictors are scored against ground truth: random, an honest single-shot LLM reading the whole transcript, and the sensor+filter loop. Over 30 episodes: loop 0.60, single-shot 0.60, random 0.233 (chance 0.20).

push	0.00	0.15	0.30	0.45	0.60	0.75	0.90
up (ρ=0.97)	.184	.198	.249	.284	.324	.364	.385
down (ρ=0.97)	.382	.405	.417	.421	.419	.422	.401
up (ρ=1.0)	.185	.198	.241	.254	.289	.323	.356
down (ρ=1.0)	.368	.379	.389	.383	.386	.376	.370

Table 2: E4 disengaging mass along the up/down ramps. Loop area 0.126 (ρ=0.97) vs. 0.115 (ρ=1.0); the hysteresis is essentially unchanged with decay removed.

The loop infers at 3× chance, but it *ties* the honest single-shot baseline: on short evidence the filter buys persistence and accumulated state, not raw accuracy. The methodological point is that a bare LLM is *not* at chance on this task, so reporting a chance-level baseline would inflate the apparent gain of the loop.

3.4 E4: closed-loop hysteresis, and the decay confound

Only this experiment probes dynamics. Closing the loop and applying an external push toward *disengaging*, ramped up ($0 \to 0.9$) then back down ($0.9 \to 0$) from the same posterior, we measure the disengaging mass at each level (Table 2). The down-ramp sits far above the up-ramp at every level (loop area 0.126): pushed to disengagement at 0.9 and released to 0, the state *stays* in the basin. But the up-ramp rises smoothly (max single-step jump 0.05): there is no discontinuous transition. So the loop shows strong **path-dependent memory** but not sharp bistability—softer than a first-order phase transition.

The decay confound—and why the obvious control is insufficient. The tracker's decay ($\rho = 0.97 < 1$) builds intrinsic inertia into the posterior, so some stickiness is mechanical rather than emergent. The natural control is a decay-free rerun: with $\rho = 1.0$ the loop area held at 0.115 (vs. 0.126) —essentially unchanged (Table 2), and the transition remains smooth under both settings. We initially read this as decisive. It is not: an accumulator at $\rho = 1.0$ never forgets, so it is path-dependent *by construction*, and "survives decay-off" cannot distinguish loop-borne memory from accumulator arithmetic. Separating the two requires a null that keeps the accumulator and removes the loop—which is E5b, below.

3.5 E5–E6: what kind of memory?

Three follow-ups decompose the E4 effect. All run at $\rho = 1.0$ on a single base model, with the push held for K turns per level (E4 corresponds to a fast ramp).

E5: ramp-rate sweep. If the hysteresis were lag—a viscous state trailing a moving input—slowing the ramp would collapse the loop area toward zero. It does the opposite (Table 3): dwelling longer at each level entrenches the state more deeply. The memory is not lag.

E5b: state-decoupled null. The decisive control. The identical ramp protocol is run with a likelihood that ignores the state entirely (no LLM in the loop), so any loop area is pure accumulator arithmetic. The null still produces areas of 0.10–0.15 with the same growth in K (Table 3): roughly two thirds of the raw E5 area is the estimator, confirming that E4's "survives decay-off" framing overstated what that rerun showed. The **loop-attributable memory** is the excess of the closed

turns per level K	1	3	6	12
closed loop (E5)	0.161	0.210	0.213	0.231
state-decoupled null (E5b)	0.101	0.135	0.142	0.153
loop-attributable excess	0.060	0.075	0.071	0.078

Table 3: Decomposing the hysteresis ($\rho = 1.0$). The null—the same accumulator with the loop removed—carries most of the raw area; the loop-attributable excess is ∼0.07 and rate-independent, growing slightly as the ramp slows rather than collapsing as a lag account would predict.

loop over the null: +0.06 to +0.08 at every K, and not shrinking as the ramp slows. Real at every rate—but about a third of the raw area.

E6: two-point basin test. Whether the residual memory amounts to true bistability—two attractors at identical input—is tested by starting the closed loop from opposite dominant modes under a clamped push of 0.45. At $N = 30$ turns the trajectories still held a ∼0.20 gap, but the lower trajectory was still climbing, so a 30-turn run cannot distinguish two attractors from slow convergence to one (the shipped `e6_bistability.json` verdict, computed at $N = 30$, should be disregarded for exactly this reason). At $N = 100$ the question resolves: both trajectories converge on a single attractor (∼0.60–0.66 disengaging mass), with tail gaps of 0.07–0.08 and still closing. **No bistability.** This also revises E4's "the state stays in the basin" reading: at long horizon it would not stay; the return is merely slow.

Revised claim. The closed loop adds modest, genuine memory: ∼0.07 of loop area beyond accumulator arithmetic, rate-independent, with initial-condition effects visible for over 100 turns—under a single attractor, with no sharp transition and no bistability. The mechanism is loop-amplified stiffening of convergence, not competing basins. Every stronger reading tested in this series, including two of our own along the way, failed its control.

4 Discussion

Across these controls the picture is consistent: each tested claim reproduces in a *true-but-softer* form. The self-state is causally load-bearing on behaviour—most clearly through the private monologue, and only once the posterior is confident. The loop is calibrated: its state predicts its own reply behaviour. Hidden-stance inference works, but against an honest baseline the loop contributes persistence rather than accuracy. And the closed loop adds genuine memory beyond its own accumulator—a modest, rate-independent excess over a state-decoupled null—under a single attractor: initial conditions stay visible for over a hundred turns, but there is no sharp transition and no bistability. Two of our own intermediate readings (that a decay-free rerun was the decisive confound control, and that a 30-turn basin gap indicated bistability) failed their own follow-up controls and are corrected above; we report them because the failure mode—an accumulator that is path-dependent by construction, and a stability criterion that passes on slow drift—will face anyone reproducing this loop.

None of this speaks to the stronger claims the architecture is often presented with. We measured structural properties of a feedback loop; whether such properties bear on any threshold or phenomenal question is outside what these experiments can address, and we make no such claim.

What we can say is narrower and, we think, useful: the loop does several real things, and it does them in a milder and better-controlled form than the original presentation asserts.

Limitations. Figures come from a small number of base models; the directions are the finding, not the magnitudes. The blind labeller in E1 has limited power on the label channel (the word-count and question-rate channels separate more cleanly). E3 uses short (three-line) episodes; longer evidence might let the filter's persistence convert into an accuracy gain over the single-shot baseline. The E5/E5b/E6 memory decomposition comes from a single base model and one push geometry; the $\sim$0.07 excess should be read as an existence result, not a constant. These are the natural next experiments.

Reproducibility

A clean-room implementation of the loop and all experiments (E1–E6, including the E5b state-decoupled null and the long-horizon E6 rerun), with the raw result files, is released under the MIT license at the project repository. The code uses only the Python standard library and targets any OpenAI-compatible endpoint. The decay-free control is a single environment variable (`SELFMODEL_RHO=1.0`).

Acknowledgements

The self-model loop architecture studied here is due to Lark Laflamme (RavenNest Scientific). This is an independent reconstruction and set of controls; it reproduces none of the original code and is built from the public descriptions. We thank the author for write-ups clear enough to rebuild from.